Linguistic Conflict Resolution: A Bilingual Perspective on Domain-General vs Domain-Specific Processes

ABSTRACT

One of the key questions in sentence processing research is how domain-general cognitive control adjustments influence the processing of ambiguous sentences. Previous work has shown a pattern known as conflict adaptation in which conflict increases cognitive control and facilitates subsequent conflict resolution (Gratton et al., 1992). Hsu and Novick (2016) reported that this adaptation effect also appears in syntactic ambiguity resolution in auditory sentence comprehension. However, it is not clear whether the magnitude of this effect varies across different task modalities or whether bilingual experience, especially differences in language typology, modulates it.

This project addresses these questions with two experiments, and it is hypothesized that bilingual individuals with typologically similar languages will exhibit stronger conflict adaptation effects, and that the Word Maze task will provide greater sensitivity to conflict-induced processing difficulty than the auditory Visual World task due to its word-by-word structure which requires a new decision at each step.

Experiment 1 replicates the Stroop to Visual World paradigm used by Hsu and Novick (2016) and Experiment 2 introduces another paradigm in interleaved design: Stroop to Word Maze task. In both experiments, it is predicted that incongruent Stroop trials will induce faster resolution of syntactic ambiguity relative to congruent trials. The study includes 2 bilingual groups with typologically similar (Spanish-English) and different (Turkish-English) languages and monolinguals as a control group (target N per group ≈40), and tests whether typological similarity between bilinguals' languages modulates conflict adaptation during ambiguous sentence processing. In addition, the study examines whether task modality influences the detectability of conflict adaptation.

The pilot phase of Experiment 2 was conducted with English monolinguals (N = 6), and in the experiment each Stroop trial was followed by a Maze sentence. Sentence-level response times were log-transformed and analysed with a linear model that included sentence type, previous Stroop condition, and their interaction. As expected for the preliminary findings from the small pilot dataset, none of the effects were significant yet (sentence type: $p = .247$, previous Stroop condition: $p = .840$, interaction: $p = .771$). Although the pilot does not show clear evidence of conflict adaptation, it confirms that the Stroop to Maze design works as intended and produces stable measures of incremental sentence processing.

Ongoing data collection on monolingual and bilingual groups includes the auditory Visual World task and the Word Maze task in interleaved design with Stroop trials. This will allow direct comparison across task modalities and test whether typological differences of bilinguals' languages modulate the role of cognitive control in ambiguous sentence comprehension.

References

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